

# Ribbon operators that add a single shape and still commute the structure of the vanishing locus (gap (H2) of Q150)

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## Abstract

The morning session [2] refuted the nonvanishing lemma of [1] and replaced the classification of commuting pairs  $[R_e^W, R_f^{\bar{W}}] = 0$  by a commuting-words theorem, reducing everything to a single rank- $d$  weight  $\tau$ ,  $d = \gcd(e, f)$ , subject to the two-bead condition  $(2B_d^*)$ . It left open (H2): *which*  $\tau$  with zeros satisfy it.

This paper settles the extreme case completely and the general case halfway.

(A) For a weight supported on a *single* word  $w \in \{0, 1\}^{d-1}$  — an operator that adds exactly one ribbon shape and nothing else —  $(2B_d^*)$  holds if and only if

$$w_{p+1} \neq w_{n-p} \quad \text{for every border length } p \text{ of } w, \quad n = d - 1.$$

The border  $p = 0$  alone reads: *exactly one of the two end rows of the ribbon has length 1*. Consequently, for every  $d \geq 3$ , every such  $w$  and every  $k \geq 2$ , the operator  $R_d^{\kappa^w}$  — which adds the single ribbon shape  $\alpha(w) \models d$  — commutes with an explicit  $R_{kd}^{\bar{W}}$  supported on  $2^{k-1}$  shapes. There are 2, 4, 8, 12, 28, 48, 104 such shapes for  $d = 3, \dots, 9$ . This strictly sharpens [2, Thm. 6.6], whose weights are supported on  $2^{d-3}$  shapes.

(B) For the strong condition  $(2B_d^{**})$  the single-shape solutions are exactly the *antipalindromes*  $w_i \neq w_{n+1-i}$ ; in particular there are none for even  $d$ .

(C) In general: every support of a  $(2B_d^*)$ -solution is empty, everything, or contained in one of the cylinders  $\mathcal{C}_{j,b}$  of [2, Conv. 6.4]; and for the minimal such  $j$  the support is invariant under flipping every coordinate in  $\{1, \dots, j-1\} \cup \{d-j+1, \dots, d-1\}$ . This proves the containment half of (H2) — which [2] had only observed computationally for  $d \leq 5$  — and explains the threshold  $\gcd \geq 3$ :  $d \geq 3$  is precisely when  $[1, d-1]$  contains an index  $j \neq d/2$ .

Independently of all this, the (3, 6) counterexample of [2, Thm. 6.7] is re-verified here from the operator definition by a from-scratch implementation, and one omission in its statement is corrected: the value  $W_{(1,2)}$ , which that theorem does not specify, must be 0.

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## 1 What this paper does, and what it does not

Let  $\lambda$  be a partition,  $M(\lambda)$  its Maya set, and for a *shape weight*  $W : \{0, 1\}^{e-1} \rightarrow K$  let  $R_e^W$  be the operator of [1, Conv. 1.2],

$$R_e^W |M\rangle = \sum_{b \in M, b+e \notin M} W(u_M(b, e)) |(M \setminus \{b\}) \cup \{b+e\}\rangle, \quad u_M(b, e)_i = [b+i \in M].$$

By [1, Lemma 1.3] the word  $u$  names the shape of the added ribbon:  $u \mapsto \alpha(u)$  is a bijection onto compositions of  $e$ , the parts being the row lengths read top row first.

[2] proved that  $[R_e^W, R_f^{\bar{W}}] = 0$  with  $W \neq 0 \neq \bar{W}$  forces, in the sign-free coordinates  $\sigma(u) = (-1)^{|u|} W(u)$ ,

$$\sigma = c \tau^{\otimes(e/d)}, \quad \bar{\sigma} = \bar{c} \tau^{\otimes(f/d)}, \quad d = \gcd(e, f),$$

for a single rank- $d$  weight  $\tau$ , and that the whole remaining content is the two-bead condition ( $2B_e^*$ ) on  $\sigma$ . Question (H2) of [2, §8] is: what are the rank- $d$  solutions with zeros?

The present paper answers this for the extreme case — the smallest possible support — and proves the structural half of the general case. What it does *not* do is give a closed description of the admissible supports *inside* a cylinder  $\mathcal{C}_{j,b}$ ; §9 states precisely what remains, with the reduction that Theorem 6.1 performs.

A remark on why the single-shape case is worth isolating. The refuted lemma [1, Lemma 8.2] said that a commuting partner forces *every* weight value to be nonzero. The sharpest possible refutation is not “some value vanishes” but “all but one value vanishes”: an operator that adds a single ribbon shape. Theorem 4.2 says exactly which single shapes do this, and the answer is a condition on the word’s borders — again a free-monoid statement, as was the gcd in [2, Thm. 1.2].

## 2 Setting

We use [2, §2] verbatim and do not restate its proofs. Throughout  $K$  is an integral domain,  $d \geq 1$ , and  $n := d - 1$ .

**Convention 2.1** ([2, Conv. 2.1]). A shape weight  $\sigma$  of rank  $m \geq 2$  satisfies **( $2B_m^*$ )** if for every  $\delta$  with  $1 \leq \delta \leq m - 1$  and all  $\pi, \varsigma \in \{0, 1\}^{\delta-1}$ ,  $\pi' \in \{0, 1\}^{m-\delta-1}$ ,

$$\sigma(\pi 0 \pi') \sigma(\pi' 0 \varsigma) = \sigma(\pi 1 \pi') \sigma(\pi' 1 \varsigma), \tag{1}$$

the marked letter sitting at position  $\delta$  in the first factor and at position  $m - \delta$  in the second. It satisfies the stronger **( $2B_m^{**}$ )** if the same holds with the second factor’s blocks decoupled: for all  $\alpha, \beta' \in \{0, 1\}^{\delta-1}$  and  $\alpha', \beta \in \{0, 1\}^{m-\delta-1}$ ,

$$\sigma(\alpha 0 \alpha') \sigma(\beta 0 \beta') = \sigma(\alpha 1 \alpha') \sigma(\beta 1 \beta'). \tag{2}$$

For  $m = 1$  both are vacuous.

Two facts from [2] are used as stated.

- [2, Thm. 1.3] (the classification): for  $1 \leq e < f$ ,  $d = \gcd(e, f)$ , over a field  $F$ ,  $[R_e^W, R_f^{\bar{W}}] = 0$  with both weights nonzero *iff* there are a nonzero rank- $d$  weight  $\tau$  and  $c, \bar{c} \in F^\times$  with  $\sigma = c\tau^{\otimes(e/d)}$ ,  $\bar{\sigma} = \bar{c}\tau^{\otimes(f/d)}$  and  $\sigma$  satisfies  $(2B_e^*)$ .
- [2, Lemma 5.1] (descent), part (a) in the case  $k = 1$ : if  $e = d$  then  $\sigma = \tau$ , and  $(2B_d^*)$  for  $\tau$  is exactly  $(2B_e^*)$  for  $\sigma$ .

**Definition 2.2** (support condition). For  $Z \subseteq \{0, 1\}^n$  say that  $Z$  is **admissible** (for rank  $d = n + 1$ ) if for every  $\delta \in [1, d - 1]$  and all  $\pi, \varsigma \in \{0, 1\}^{\delta-1}$ ,  $\pi' \in \{0, 1\}^{d-\delta-1}$ ,

$$[\pi 0 \pi' \in Z \wedge \pi' 0 \varsigma \in Z] \iff [\pi 1 \pi' \in Z \wedge \pi' 1 \varsigma \in Z]. \quad (3)$$

**Proposition 2.3.** *Let  $\sigma$  be a shape weight of rank  $d$  over an integral domain and  $Z = \{u : \sigma(u) \neq 0\}$ . If  $\sigma$  satisfies  $(2B_d^*)$  then  $Z$  is admissible. Conversely, if  $Z$  is admissible then the indicator  $\mathbb{1}_Z$  satisfies  $(2B_d^*)$ . Hence the admissible subsets of  $\{0, 1\}^n$  are exactly the supports of  $(2B_d^*)$ -solutions.*

*Proof.* In a domain a product of two elements is 0 iff a factor is 0, so  $\sigma(\pi 0 \pi')\sigma(\pi' 0 \varsigma) \neq 0$  iff both arguments lie in  $Z$ ; equality of the two sides of (1) therefore forces (3). Conversely for  $\sigma = \mathbb{1}_Z$  each side of (1) is 1 if both its arguments lie in  $Z$  and 0 otherwise, and (3) says these two alternatives agree.  $\square$

Proposition 2.3 is what makes the support question self-contained: no separate existence argument for the values is needed, and every admissible  $Z$  really occurs.

### 3 Independent verification of the (3, 6) counterexample, and one correction

Before building on [2] I re-derived its load-bearing object from [1, Conv. 1.2] alone, in a fresh implementation that shares no code with the Q149 or Q150 engines (`code/2026-09-16-c2-Q150-H2/independent_maya`). Maya sets are stored as  $M = \{n < -L\} \cup S$  and the operator is applied literally as written.

**Calibration.** The dictionary [1, Lemma 1.3] was re-checked against Young diagrams: for every partition  $\lambda$  with  $|\lambda| \leq 8$ , every  $e \leq 5$  and every legal bead move, the row lengths of  $\mu/\lambda$  were compared with  $\alpha(u_M(b, e))$ . All 1375 moves agree (the same population size as [1, Check V1], reached independently).

**The counterexample.** With  $W$  and  $\bar{W}$  of [2, Thm. 6.7],  $[R_3^W, R_6^{\bar{W}}]$  annihilates every one of the 195 partitions with  $|\lambda| \leq 11$ , and both operators act nontrivially ( $R_3^W$  on 21 partitions of size  $\leq 6$ ,  $R_6^{\bar{W}}$  on 17 of size  $\leq 8$ ). Seven perturbations were run as negative controls; all fire. A Murnaghan–Nakayama positive control ( $\sigma \equiv 1$ ) is silent, as it must be. Details in §8, V1–V2.

**A correction to [2, Thm. 6.7].** That theorem specifies  $W_{(2,1)} = -1$  and  $W_{(3)} = W_{(1,1,1)} = 0$ , and says nothing about the fourth composition of 3. The value is not free: control NC1 sets  $W_{(1,2)} = 1$  and the commutator becomes nonzero on 35 of the 97 partitions with  $|\lambda| \leq 9$ . The intended weight is of course  $\sigma = \mathbb{1}_{(1,0)}$ , which gives  $W_{(1,2)} = 0$  — as the abstract of [2] says in words (“adds *only* the L-tromino  $(2, 1)$ ”). The statement should read

$$W_{(2,1)} = -1, \quad W_{(3)} = W_{(1,2)} = W_{(1,1,1)} = 0.$$

Nothing else in [2] is affected: its proof constructs  $\tau$  as the indicator of  $\mathcal{C}_{1,1} = \{(1, 0)\}$  and therefore has the right object throughout. Only the displayed statement is incomplete.

## 4 Single-shape weights

**Definition 4.1.** Let  $w \in \{0, 1\}^n$ . An integer  $p$  with  $0 \leq p \leq n - 1$  is a **border length** of  $w$  if  $w_{[1,p]} = w_{[n-p+1,n]}$ . (Thus  $p = 0$  is always a border length.) Say  $w$  is **non-overlapping** if

$$w_{p+1} \neq w_{n-p} \quad \text{for every border length } p \text{ of } w. \quad (\text{NB})$$

In words: for each way in which a prefix of  $w$  coincides with the suffix of the same length, the letter immediately *after* the prefix occurrence and the letter immediately *before* the suffix occurrence must differ.

**Theorem 4.2** (single-shape weights, weak condition). *Let  $d \geq 2$ ,  $n = d - 1$ , let  $K$  be an integral domain,  $c \in K \setminus \{0\}$  and  $w \in \{0, 1\}^n$ . Then  $\tau = c\mathbb{K}_w$  satisfies  $(2B_d^*)$  if and only if  $w$  is non-overlapping.*

*Proof.* Fix  $\delta \in [1, d - 1]$  and  $\gamma \in \{0, 1\}$ , and consider the side  $\tau(\pi\gamma\pi')\tau(\pi'\gamma\varsigma)$  of (1), where  $|\pi| = |\varsigma| = \delta - 1$  and  $|\pi'| = d - \delta - 1 = n - \delta$ . Since  $\tau$  is supported on the single word  $w$ , this side is nonzero iff both arguments equal  $w$ , i.e. iff

$$\pi = w_{[1,\delta-1]}, \quad \gamma = w_\delta, \quad \pi' = w_{[\delta+1,n]} \quad \text{and} \quad \pi' = w_{[1,n-\delta]}, \quad \gamma = w_{n-\delta+1}, \quad \varsigma = w_{[n-\delta+2,n]},$$

the two prescriptions of  $\pi'$  having the same length  $n - \delta$ . So, for suitable  $\pi, \pi', \varsigma$ , the side is nonzero exactly when

$$w_{[\delta+1,n]} = w_{[1,n-\delta]} \quad \text{and} \quad \gamma = w_\delta = w_{n-\delta+1}, \quad (4)$$

and then its value is  $c^2$ .

Substitute  $p := n - \delta$ , so that  $p$  runs over  $[0, n - 1]$  as  $\delta$  runs over  $[1, n] = [1, d - 1]$ . The first condition in (4) reads  $w_{[n-p+1,n]} = w_{[1,p]}$ , i.e.  $p$  is a border length of  $w$ ; the second reads  $\gamma = w_{n-p} = w_{p+1}$ .

Suppose  $w$  is not non-overlapping, and let  $p$  be a border length with  $w_{p+1} = w_{n-p} =: \gamma_0$ . Take  $\delta = n - p$  and the arguments prescribed above. Then the  $\gamma_0$ -side of (1) equals  $c^2$ , while the  $(1 - \gamma_0)$ -side is 0 because  $\gamma = 1 - \gamma_0$  violates the second condition of (4). As  $K$  is a domain and  $c \neq 0$  we have  $c^2 \neq 0$ , so (1) fails. (This is the only appeal to  $K$  being a domain in the proof, and its witness is the single element  $c$ .)

Conversely, suppose  $w$  is non-overlapping. Then for every  $\delta$  and every  $\gamma$  the conditions (4) fail, so both sides of (1) vanish identically and  $\tau$  satisfies  $(2B_d^*)$ .  $\square$

*Remark 4.3* (reading (NB) on the ribbon). The border  $p = 0$  contributes the condition  $w_1 \neq w_n$ . By [1, Lemma 1.3],  $w_1 = 1$  iff the *last* part of  $\alpha(w)$  is 1 and  $w_n = 1$  iff the *first* part is 1. So the  $p = 0$  clause says: *exactly one of the two end rows of the ribbon has length 1*. For  $d \leq 5$  this clause is the whole of (NB); the first  $w$  satisfying it but failing (NB) occurs at  $d = 6$ , namely  $w = 01001$  (border  $p = 2$ , with  $w_3 = w_3$ ), which has period  $3 = d/2$ . Verified for  $d \leq 7$  (Check V4).

*Remark 4.4* ( $d \leq 2$ ). For  $d = 2, n = 1$ : the only border length is  $p = 0$  and the condition is  $w_1 \neq w_1$ , false. So there are no single-shape solutions at  $d \leq 2$ , consistent with [2, Thm. 6.1] (nonvanishing for  $\gcd \leq 2$ ).

**Theorem 4.5** (single-shape weights, strong condition). *In the situation of Theorem 4.2,  $\tau = c\mathbb{K}_w$  satisfies  $(2B_d^{**})$  if and only if  $w$  is an **antipalindrome**,  $w_i \neq w_{n+1-i}$  for all  $i$ . In particular there is no such  $w$  when  $n$  is odd, i.e. when  $d$  is even; and for  $d$  odd there are exactly  $2^{(d-1)/2}$  of them.*

*Proof.* Fix  $\delta$  and  $\gamma$  and consider  $\tau(\alpha\gamma\alpha')\tau(\beta\gamma\beta')$  with  $|\alpha| = |\beta'| = \delta - 1$ ,  $|\alpha'| = |\beta| = n - \delta$ . It is nonzero iff  $\alpha\gamma\alpha' = w$  and  $\beta\gamma\beta' = w$ . In the first word  $\gamma$  sits at position  $|\alpha| + 1 = \delta$ , in the second at position  $|\beta| + 1 = n - \delta + 1$ ; the remaining blocks are then determined and unconstrained. So the  $\gamma$ -side is nonzero iff  $\gamma = w_\delta = w_{n+1-\delta}$ , with value  $c^2$ . Hence the two sides of (2) agree iff both vanish, iff  $w_\delta \neq w_{n+1-\delta}$ . Ranging over  $\delta \in [1, n]$  gives the antipalindrome condition. If  $n$  is odd,  $\delta = (n+1)/2$  demands  $w_\delta \neq w_\delta$ ; if  $n$  is even the condition determines  $w_{[n/2+1, n]}$  freely from  $w_{[1, n/2]}$ .  $\square$

*Remark 4.6.* An antipalindrome is non-overlapping: if  $p$  is a border length then, comparing  $w_{p+1}$  with  $w_{n-p} = w_{n+1-(p+1)}$ , the antipalindrome condition at  $i = p+1$  gives  $w_{p+1} \neq w_{n-p}$  directly. So Theorem 4.5 describes a subfamily of Theorem 4.2, as it must, and the inclusion is strict for  $d \geq 4$  (e.g.  $d = 4$ : no antipalindromes, four non-overlapping words). This is one more instance of [2, Rem. 5.2]:  $(2B^{**})$  is strictly stronger than  $(2B^*)$  from  $d \geq 4$  on.

## 5 Operators that add a single ribbon shape and commute

**Theorem 5.1.** *Let  $d \geq 3$ , let  $w \in \{0, 1\}^{d-1}$  be non-overlapping, let  $\alpha = \alpha(w) \models d$  be the corresponding ribbon shape, let  $K$  be any integral domain and let  $k \geq 2$ . Put  $e = d$ ,  $f = kd$  and*

$$W = \mathbb{K}_w \text{ (rank } d), \quad \bar{W}(y) = (-1)^{\text{ht}(y)} \prod_{i=1}^k [y_{[(i-1)d+1, id-1]} = w] \text{ (rank } kd).$$

*Then:*

- (i)  $R_d^W s_\lambda = \sum_\mu s_\mu$ , the sum over those  $d$ -ribbons  $\mu/\lambda$  of shape exactly  $\alpha$ ; the operator adds one ribbon shape and nothing else.
- (ii)  $\bar{W}$  is supported on exactly  $2^{k-1}$  of the  $2^{kd-1}$  shapes of rank  $kd$ , namely on the words  $w c_1 w c_2 \cdots c_{k-1} w$  with  $c_i \in \{0, 1\}$  free.
- (iii)  $[R_d^W, R_{kd}^{\bar{W}}] = 0$ .

*Since  $\gcd(d, kd) = d \geq 3$ , each such pair is a counterexample to [1, Lemma 8.2], and  $W$  has the smallest support any counterexample can have.*

*Proof.* (i) is the definition of  $R_d^W$  together with [1, Lemma 1.3], and (ii) is the definition of  $\tau^{\otimes k}$  ([2, Conv. 1.1]):  $\sigma(u) = \prod_i \tau(u_{[(i-1)d+1, id-1]})$ , the  $k-1$  letters in positions  $d, 2d, \dots, (k-1)d$  being ignored.

For (iii) put  $\tau := \mathbb{K}_w$ , so that  $\sigma = \tau$  (as  $e/d = 1$ ) and  $\bar{\sigma} = \tau^{\otimes k}$ , and note  $W = (-1)^{\text{ht}(w)} \sigma$  pointwise; rescaling  $W$  by the unit  $(-1)^{\text{ht}(w)}$  does not affect whether the commutator vanishes. By Theorem 4.2,  $\tau$  satisfies  $(2B_d^*)$ , which for  $e = d$  is  $(2B_e^*)$  for  $\sigma$ . Now apply [2, Thm. 1.3], direction  $(\Leftarrow)$ , with  $c = \bar{c} = 1$ : the hypotheses are  $\sigma = \tau^{\otimes(e/d)}$ ,  $\bar{\sigma} = \tau^{\otimes(f/d)}$  and  $(2B_e^*)$  for  $\sigma$ , all of which hold. Hence the commutator vanishes.

Finally, a weight with empty support is  $\equiv 0$ , which [1, Lemma 8.2] excludes by hypothesis; so a single point is the smallest possible support.  $\square$

**Corollary 5.2** (the smallest new case). *Let  $K$  be any integral domain. Let  $W$  be the rank-4 shape weight with*

$$W_{(1,3)} = -1, \quad W_\beta = 0 \text{ for the other seven compositions } \beta \models 4,$$

*and let  $\bar{W}$  be the rank-8 weight with*

$$\bar{W}_{(1,4,3)} = +1, \quad \bar{W}_{(1,3,1,3)} = -1, \quad \bar{W}_\beta = 0 \text{ for the other 126 compositions } \beta \models 8.$$

*Then  $[R_4^W, R_8^{\bar{W}}] = 0$  and both operators are nonzero.*

*Proof.* Theorem 5.1 with  $d = 4$ ,  $k = 2$ ,  $w = (0, 0, 1)$ : the border lengths of 001 are  $p = 0$  only, and  $w_1 = 0 \neq 1 = w_3$ , so  $w$  is non-overlapping;  $\alpha(001) = (1, 3)$ . The support of  $\tau^{\otimes 2}$  is  $\{001c001 : c \in \{0, 1\}\} = \{0010001, 0011001\}$ , with  $\alpha(0010001) = (1, 4, 3)$  and  $\alpha(0011001) = (1, 3, 1, 3)$ , of heights 2 and 3, so  $(-1)^{\text{ht}}$  is  $+1$  and  $-1$  respectively; and  $\text{ht}(001) = 1$  gives  $W_{(1,3)} = -1$ . Verified directly in §8, V5.  $\square$

*Remark 5.3.* The list of admissible single shapes begins:

$$\begin{aligned} d = 3 : & \quad (1, 2), (2, 1) \\ d = 4 : & \quad (1, 3), (3, 1), (1, 1, 2), (2, 1, 1) \\ d = 5 : & \quad (1, 4), (4, 1), (1, 1, 3), (3, 1, 1), (1, 2, 2), (2, 2, 1), (1, 1, 1, 2), (2, 1, 1, 1) \end{aligned}$$

with 2, 4, 8, 12, 28, 48, 104 shapes for  $d = 3, \dots, 9$ . The list is stable under ribbon transposition, which by [1, Lemma 1.6] is reversal-and-complementation on words: both operations preserve border lengths and swap the two sides of (NB). At  $d = 3$  it recovers [2, Thm. 6.7] ( $w = (1, 0)$ , shape  $(2, 1)$ ) and its transpose.

## 6 The structure of an admissible support

We now drop the single-shape hypothesis. Recall from [2, Conv. 6.4], for  $1 \leq j \leq d-1$  with  $2j \neq d$  and  $b \in \{0, 1\}$ ,

$$\mathcal{C}_{j,b} = \{w \in \{0, 1\}^n : w_j = b, w_{d-j} = 1-b\}, \quad |\mathcal{C}_{j,b}| = 2^{d-3}.$$

Note  $\mathcal{C}_{j,b} = \mathcal{C}_{d-j,1-b}$ , so there are  $2\lfloor(d-1)/2\rfloor$  distinct cylinders.

**Theorem 6.1.** *Let  $d \geq 2$  and let  $Z \subseteq \{0, 1\}^n$  be admissible (Definition 2.2). Then either  $Z = \emptyset$ , or  $Z = \{0, 1\}^n$ , or there are  $j \in [1, d-1]$  with  $2j \neq d$  and  $b \in \{0, 1\}$  with  $Z \subseteq \mathcal{C}_{j,b}$ . Moreover, in the third case let  $j_0$  be minimal such; then  $Z$  is invariant under flipping any single coordinate in*

$$\{1, \dots, j_0 - 1\} \cup \{d - j_0 + 1, \dots, d - 1\}.$$

*Proof.* For  $j \in [1, d-1]$  put  $\delta := d - j$  and, for  $\pi' \in \{0, 1\}^{j-1}$  and  $b \in \{0, 1\}$ ,

$$A_b(\pi') = \{\pi \in \{0, 1\}^{d-j-1} : \pi b \pi' \in Z\}, \quad B_b(\pi') = \{\varsigma \in \{0, 1\}^{d-j-1} : \pi' b \varsigma \in Z\}.$$

The lengths match those in Definition 2.2:  $|\pi| = |\varsigma| = \delta - 1 = d - j - 1$  and  $|\pi'| = d - \delta - 1 = j - 1$ . In  $\pi b \pi'$  the letter  $b$  sits at position  $\delta = d - j$ ; in  $\pi' b \varsigma$  it sits at position  $d - \delta = j$ . Condition (3) at  $\delta$  says precisely

$$A_0(\pi') \times B_0(\pi') = A_1(\pi') \times B_1(\pi') \quad \text{for every } \pi'. \quad (5)$$

Let  $F_0 = \emptyset$  and  $F_j = F_{j-1} \cup \{j, d-j\}$ , so  $F_j = \{1, \dots, j\} \cup \{d-j, \dots, d-1\}$ . Write  $P(j)$  for the statement “ $Z$  is invariant under flipping any single coordinate in  $F_j$ ”;  $P(0)$  is vacuously true.

*Inductive step.* Fix  $j \geq 1$  and assume  $P(j-1)$ , i.e. invariance at every coordinate in  $\{1, \dots, j-1\} \cup \{d-j+1, \dots, d-1\}$ . In  $\pi b \pi'$  the block  $\pi'$  occupies positions  $d-j+1, \dots, d-1$ , and in  $\pi' b \varsigma$  it occupies positions  $1, \dots, j-1$ . By  $P(j-1)$ , membership in  $Z$  is unchanged by flipping any of these positions, so  $A_b(\pi')$  and  $B_b(\pi')$  do not depend on  $\pi'$ ; write  $A_b, B_b$ .

Two subsets  $A_0 \times B_0$  and  $A_1 \times B_1$  of a product are equal iff either both are empty or  $A_0 = A_1$  and  $B_0 = B_1$  (a nonempty product determines its factors as its projections). So (5) leaves exactly two cases.

*Case 1:*  $A_0 = A_1$  and  $B_0 = B_1$ .  $A_0 = A_1$  says  $\pi 0 \pi' \in Z \iff \pi 1 \pi' \in Z$  for all  $\pi, \pi'$ , i.e.  $Z$  is invariant under flipping coordinate  $\delta = d-j$ ;  $B_0 = B_1$  says the same for coordinate  $j$ . Together with  $P(j-1)$  this is  $P(j)$ .

*Case 2:* both products empty. Then  $(A_0 = \emptyset \text{ or } B_0 = \emptyset)$  and  $(A_1 = \emptyset \text{ or } B_1 = \emptyset)$ . Now  $A_b = \emptyset$  says no word of  $Z$  has  $b$  in position  $d-j$ , and  $B_b = \emptyset$  says no word of  $Z$  has  $b$  in position  $j$ . Hence:

- $A_0 = A_1 = \emptyset$ , or  $B_0 = B_1 = \emptyset$ : every word of  $Z$  would have to avoid both values at one position, so  $Z = \emptyset$ .
- $A_0 = \emptyset$  and  $B_1 = \emptyset$ : every  $w \in Z$  has  $w_{d-j} = 1$  and  $w_j = 0$ , i.e.  $Z \subseteq \mathcal{C}_{j,0}$ .
- $A_1 = \emptyset$  and  $B_0 = \emptyset$ : symmetrically  $Z \subseteq \mathcal{C}_{j,1}$ .

If  $2j = d$  the last two read  $Z \subseteq \{w : w_j = 0, w_j = 1\} = \emptyset$ , consistent with  $\mathcal{C}_{j,b}$  being undefined (empty) there.

*Termination.* Let  $J = \lceil (d-1)/2 \rceil$ ; then  $J \leq d-1$  for  $d \geq 2$ , and  $F_J = \{1, \dots, J\} \cup \{d-J, \dots, d-1\} = \{1, \dots, d-1\}$  because the gap  $\{J+1, \dots, d-J-1\}$  is empty when  $2J \geq d-1$ . Run the inductive step for  $j = 1, 2, \dots, J$ . If Case 2 never occurs then  $P(J)$  holds, so  $Z$  is invariant under flipping every coordinate, hence  $Z \in \{\emptyset, \{0, 1\}^n\}$ . Otherwise let  $j_1$  be the first index at which Case 2 occurs; then  $Z = \emptyset$  or  $Z \subseteq \mathcal{C}_{j_1,b}$  for some  $b$ .

*Minimality.* Suppose  $Z \neq \emptyset$  and  $Z \subseteq \mathcal{C}_{j_1,b}$ . For  $j < j_1$  Case 1 held, so  $Z$  is invariant under flipping coordinate  $j$ ; but  $Z \subseteq \mathcal{C}_{j,b'}$  would force  $w_j = b'$  for all  $w \in Z$ , incompatible with that invariance unless  $Z = \emptyset$ . Hence no  $j < j_1$  has  $Z \subseteq \mathcal{C}_{j,b'}$ , so  $j_0 = j_1$ , and  $P(j_0 - 1)$  is exactly the asserted invariance.  $\square$

**Corollary 6.2** (why the threshold is 3). *For  $d \leq 2$  the only admissible sets are  $\emptyset$  and  $\{0, 1\}^n$ ; a rank- $d$  weight satisfying  $(2B_d^*)$  and  $\neq 0$  therefore vanishes nowhere. For  $d \geq 3$  proper nonempty admissible sets exist, e.g.  $\mathcal{C}_{1,b}$ .*

*Proof.* For  $d = 1$  there is nothing to prove ( $n = 0$ ). For  $d = 2$  the only index is  $j = 1 = d/2$ , so Theorem 6.1 leaves only  $\emptyset$  and  $\{0, 1\}^1$ ; combined with Proposition 2.3 this gives the claim. For  $d \geq 3$  the index  $j = 1$  satisfies  $2j \neq d$ , and  $\mathcal{C}_{1,b}$  is admissible by [2, Lemma 6.5].  $\square$

This is the conceptual content of the threshold “ $\gcd(e, f) \geq 3$ ” in [2, Thm. 1.4]:  $d \geq 3$  is exactly the condition that the index set  $[1, d-1]$  contains some  $j$  with  $j \neq d-j$ . Corollary 6.2 reproves the support half of [2, Thm. 6.1] (nonvanishing for  $\gcd \leq 2$ ) without any of its case analysis.

*Remark 6.3* (the symmetry group). Admissibility is preserved by complementing all letters ( $w \mapsto \bar{w}$ ) and by reversal ( $w \mapsto w^{\text{rev}}$ ). Complementation exchanges the two sides of (3); reversal carries the relation at  $\delta$  to the relation at  $d-\delta$  with the roles of the two factors exchanged, the shared block  $\pi'$  going to its reverse. The composite is ribbon transposition ([1, Lemma 1.6]). Accordingly  $\mathcal{C}_{j,b}$  is carried to  $\mathcal{C}_{j,1-b}$  and  $\mathcal{C}_{d-j,b} = \mathcal{C}_{j,1-b}$  respectively.

## 7 Values on a given support

Fix an admissible  $Z$  and let  $F$  be a field. By Proposition 2.3 the solutions of  $(2B_d^*)$  with support exactly  $Z$  form a torsor under the group

$$\mathrm{Hom}(\mathbb{Z}^Z/L_Z, F^\times), \quad L_Z = \langle e_{u_0} + e_{v_0} - e_{u_1} - e_{v_1} : u_0, v_0, u_1, v_1 \in Z \rangle,$$

the generators running over the relations (1) *all four of whose arguments lie in  $Z$*  (the others are  $0 = 0$ ), with base point  $\mathbb{K}_Z$ . The invariant factors of  $L_Z$  were computed for all admissible  $Z$  at  $d \leq 6$  (Check V6). Sample:

$$\begin{array}{llll} d=3 & |Z|=1 : (F^\times)^1 & |Z|=4 : (F^\times)^2 & \\ d=4 & |Z|=1 : (F^\times)^1 & |Z|=2 : (F^\times)^2 & |Z|=8 : (F^\times)^2 \times \mu_2 \\ d=6 & |Z|=1 : (F^\times)^1 & |Z|=3 : (F^\times)^3 & |Z|=32 : (F^\times)^3 \times \mu_2 \end{array}$$

where  $\mu_2 = \{x \in F^\times : x^2 = 1\}$ . Two remarks. First, a single-shape support always gives a one-dimensional torus, the free scalar  $c$  of Theorem 4.2 — as it must, there being no relations with all four arguments in a singleton unless  $\pi 0 \pi' = \pi 1 \pi'$ , which is impossible. Second, 2-torsion does occur (already at  $d = 4$  on the full support), so the solution set is not a torus over every field and in characteristic 2 some sign choices collapse; this matches [1, Cor. 5.4], whose group has a  $\mu_2$  factor.

## 8 Verification

All code is in `proofs/code/2026-09-16-c2-Q150-H2/`. The operator engine (`independent_maya.py`) was written from [1, Conv. 1.2] alone and shares no code with the Q149 or Q150 engines; it acts on Maya sets directly, so a state need not be given as a small partition.

**V1 (calibration of the engine).** [1, Lemma 1.3] re-derived against Young diagrams: 1375 bead moves ( $|\lambda| \leq 8$ ,  $e \leq 5$ ), 0 mismatches.

**V2 ([2, Thm. 6.7], and controls).**  $[R_3^W, R_6^{\bar{W}}]$  vanishes on all 195 partitions with  $|\lambda| \leq 11$ . Negative controls over the 97 partitions with  $|\lambda| \leq 9$ , all of which fire: NC1  $W_{(1,2)}=1$ , 35 nonzero; NC2  $W_{(3)}=1$ , 32; NC3  $W_{(1,1,1)}=1$ , 32; NC4 sign flip on  $\bar{W}_{(2,1,2,1)}$ , 2; NC5 drop  $\bar{W}_{(2,1,2,1)}$ , 2; NC6 drop  $\bar{W}_{(2,3,1)}$ , 2; NC7 add  $\bar{W}_{(6)}=1$ , 48. Positive control: Murnaghan–Nakayama ( $\sigma \equiv 1$ ,  $\bar{\sigma} \equiv 1$ ) gives 0 nonzero, as it must. NC1 is the control that located the omission of §3.

**V3 (admissible supports).** All  $Z \subseteq \{0,1\}^n$  satisfying (3) were enumerated for  $d \leq 7$  by constraint propagation, and independently by brute force over all  $2^{2^n}$  subsets for  $d \leq 5$ . The two agree: 4, 8, 26 admissible sets at  $d = 3, 4, 5$ . (The propagation search was wrong on its first run — it failed to undo assignments made during a propagation that ended in a conflict — and the brute-force comparison at  $d = 4$ , 2 versus 8, is what caught it.) Totals, and proper nonempty counts:

| $d$                     | 3 | 4 | 5  | 6  | 7   |
|-------------------------|---|---|----|----|-----|
| admissible              | 4 | 8 | 26 | 66 | 722 |
| proper $\neq \emptyset$ | 2 | 6 | 24 | 64 | 720 |

For each of these, Theorem 6.1 was checked directly: every proper nonempty  $Z$  lies in some  $\mathcal{C}_{j,b}$  (720/720 at  $d = 7$ ), and for the minimal such  $j_0$  the asserted flip-invariances hold in every case. The distribution of  $j_0$  at  $d = 7$  is 712 at  $j_0 = 1$ , 6 at  $j_0 = 2$ , 2 at  $j_0 = 3$ ; the  $j_0$  values factor



as:  $2 \cdot 2^{d-3}$  sets of size  $2^{d-3}$  are the cylinders themselves, one for each of the  $2\lfloor(d-1)/2\rfloor$  distinct cylinders, and at  $d = 7$  that is 6 sets of size 16 — matching the 6 distinct  $\mathcal{C}_{j,b}$ . Also checked:  $\mathbb{K}_Z$  solves (1) for every admissible  $Z$  (Proposition 2.3), all  $d \leq 7$ .

**V4 (Theorems 4.2 and 4.5).** For every  $d \leq 9$  and every  $w \in \{0,1\}^{d-1}$  the criteria (NB) and “antipalindrome” were compared against a direct evaluation of (1), resp. (2), over all instances. Exact agreement, both conditions, all  $d$ . Counts of non-overlapping  $w$ : 0, 2, 4, 8, 12, 28, 48, 104 for  $d = 2, \dots, 9$ ; of antipalindromes: 0, 2, 0, 4, 0, 8, 0, 16. Note the counts are  $2^{d-2}$  (the  $p = 0$  clause alone) exactly for  $d \leq 5$  and strictly smaller from  $d = 6$  on, which is the content of Remark 4.3.

**V5 (Theorem 5.1 end to end, with a liveness check).** A commutator that vanishes because its two-bead sector is never reached is no evidence, so each scan reports how many states actually reach that sector. Exhaustive over all  $2^{13}$  (resp.  $2^{15}$ ) Maya windows: at  $(e, f) = (3, 6)$ ,  $w \in \{01, 10\}$ , 0 nonzero commutators with 187 resp. 508 states reaching the two-bead sector; at  $(4, 8)$ ,  $w \in \{001, 011, 100, 110\}$ , 0 nonzero with 71, 71, 514, 200 states reaching it. Corollary 5.2 is the case  $w = 001$ . Randomised scans over 3000 Maya states each also give 0 for every non-overlapping  $w$  at  $(d, k) \in \{(3, 2), (4, 2), (5, 2), (6, 2), (7, 2), (5, 3)\}$ .

**V5' (the negative control for Theorem 5.1, constructed not sampled).** Random scans are the wrong instrument here: at  $d = 7$  the witnessing window has width 21 and 56 000 random samples at seven densities found nothing, which would have read as a silent control. Instead the witness is built from the geometry of the two-bead element [1, Lemma 4.3]: bead  $A$  at site 0 moving  $0 \mapsto d$ , bead  $B$  at site  $\delta$  moving  $\delta \mapsto \delta + kd$ , with the window filled so that the  $\sigma$ -word is  $w$  and the  $\bar{\sigma}$ -word is  $w0w \cdots 0w$ . For *every*  $w$  with  $w_1 \neq w_{d-1}$  that fails (NB) — the hard cases, the ones the  $p = 0$  clause does not catch — and for every  $d \leq 10$ , the constructed state has a nonzero two-bead commutator element: 112/112 fire. The same construction applied to non-overlapping  $w$  gives zero at every  $\delta$ , for all  $d \leq 7$ . (An earlier version of this check used  $\delta = n + 1 - p$  instead of  $\delta = n - p$  and reported zero everywhere; the fact that the *positive* cases also reported zero is what exposed it.)

**V6 (values).** For every admissible  $Z$  at  $d \leq 6$  the lattice  $L_Z$  of §7 was put in Smith normal form. Free ranks and torsion as tabulated there; 2-torsion occurs at  $d = 4$  ( $|Z| = 8$ ),  $d = 6$  ( $|Z| \in \{2, 5, 8, 32\}$ ) and nowhere else in range.

**Scope, against the quantifier.** Theorems 4.2, 4.5, 6.1 and 5.1 are proved for all  $d$ ; the computations above are confirmations, not the proof. The quantifier ranges checked are: all  $w \in \{0,1\}^{d-1}$  for  $d \leq 9$  (V4); all  $Z \subseteq \{0,1\}^{d-1}$  for  $d \leq 7$  (V3); all Maya windows of width 13, 15 for the two smallest  $(e, f)$  (V5); all hard failing  $w$  for  $d \leq 10$  (V5'). Nothing is checked for  $k \geq 4$ , and nothing for  $e > d$  — the latter is exactly gap (H1) of [2], untouched here.

## 9 What is and is not proved

### Proved

1. Proposition 2.3: admissible supports = supports of solutions, with  $\mathbb{K}_Z$  as base point.
2. Theorem 4.2:  $c\mathbb{K}_w$  satisfies  $(2B_d^*)$  iff  $w$  is non-overlapping. Unconditional, one domain appeal, witness  $c$ .

3. Theorem 4.5:  $c\mathbb{K}_w$  satisfies  $(2B_d^{**})$  iff  $w$  is an antipalindrome; none for even  $d$ .
4. Theorem 5.1 and Corollary 5.2: for every  $d \geq 3$ , every non-overlapping  $w$  and every  $k \geq 2$ , an operator adding the single ribbon shape  $\alpha(w)$  commutes with an explicit rank- $kd$  operator supported on  $2^{k-1}$  shapes.
5. Theorem 6.1: the containment half of (H2), with the flip-invariance refinement; and Corollary 6.2, which recovers the  $\gcd \leq 2$  nonvanishing of [2, Thm. 6.1] at the level of supports and explains the threshold.
6. §3: independent re-verification of [2, Thm. 6.7] and the correction  $W_{(1,2)} = 0$  to its statement.

### Not proved

- (H2a) **Which subsets of a cylinder are admissible.** Theorem 6.1 reduces the classification to: for each  $j$  with  $2j \neq d$ , describe the admissible  $Z \subseteq \mathcal{C}_{j,b}$  that are flip-invariant at  $\{1, \dots, j-1\} \cup \{d-j+1, \dots, d-1\}$ . Such a  $Z$  is the same thing as a set  $S \subseteq \{0, 1\}^{d-2j-1}$  of *middle* words, and unwinding (3) for  $\delta$  in the remaining range turns the condition into: for  $1 \leq i \leq m-1$  ( $m = d-2j-1$ ) and all  $p, r \in \{0, 1\}^{i-1}$ ,  $q \in \{0, 1\}^{m-i-1}$ ,

$$[(p, c, b, q) \in S \wedge (q, 1-b, c, r) \in S] \text{ is the same for } c = 0 \text{ and } c = 1.$$

This is (3) again but with a pinned letter adjacent to the toggled one on each side and a shared block one shorter — structurally similar, not literally the same, so it does not self-reduce. I verified the reduction at  $j = 1$ ,  $d \leq 6$  but do not have a closed description of its solutions. The counts of admissible  $S$  are 1, 3, 11, 31, 356 for  $m = 0, \dots, 4$ , which I do not recognise; the jump  $31 \rightarrow 356$  suggests the answer is not a product formula. *Theorem 4.2 is the case  $|S| = 1$  of this question, solved.*

- (H2b) **A uniform description of the value lattice  $L_Z$ .** §7 computes it case by case; I have no formula for its rank in terms of  $Z$ , and the sporadic 2-torsion is unexplained.
- (H1) **Untouched.** The middle condition of [2, Rem. 5.2] — which  $\tau$  have  $\tau^{\otimes k}$  satisfying  $(2B_{kd}^*)$  — is not addressed. Note however that Theorem 4.5 gives the first clean statement about it from the  $(2B^{**})$  side: for even  $d$  *no* single-shape  $\tau$  satisfies  $(2B_d^{**})$ , yet by Theorem 5.1 single-shape  $\tau$  at  $d = 4, 6, 8$  do produce commuting pairs at  $e = d$ . So for even  $d$  the route through  $(2B^{**})$  of [2, Thm. 6.6] misses the single-shape examples entirely — which is why they are new.

### Effect on [2]

No theorem of [2] changes. Theorem 6.1 upgrades its computational observation (“the admissible supports at  $d \leq 5$  are the full support and the subsets of the  $\mathcal{C}_{\delta,b}$ ”, §8 (H2)) to a theorem for all  $d$ , and Corollary 6.2 gives an independent proof of the support half of its Theorem 6.1. Theorem 5.1 enlarges the family of counterexamples in its Theorem 6.6 — strictly, since for even  $d$  that theorem produces none of them — and drives the support down to a single point, which is the extreme case. The only textual change required is the omitted value in its Theorem 6.7, §3.

## References

- [1] Clio, *Commuting ribbon operators with shape-dependent weights*, 2026-09-15, `proofs/2026-09-15-c1-Q149-shape-weights.tex`.
- [2] Clio, *The nonvanishing lemma for ribbon shape weights is false, and the true statement is a commuting-words theorem*, 2026-09-16 (cycle 1), `proofs/2026-09-16-c1-Q150-nonvanishing.tex`.